Dylan Kaing

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EDUCATION

University of Pittsburgh – School of Computing and Information Aug 2022 - Dec 2025
B.S., Computer Science

- GPA: 3.59
- Coursework: Algorithms & Data Structures, Operating Systems, Applied Cryptography & Network Security, Database Management Systems, Cloud Computing, Computer Graphics

EXPERIENCE

University of Pittsburgh × CGI Partners Aug 2025 - Dec 2025
Full-Stack Developer – CGI Web Application

- Partnered with CGI to design and develop a web application that streamlined internal corporate workflows, improving operational efficiency for end users.
- Built a responsive front end with Angular and TypeScript, backed by a SpringBoot REST API with Swagger-documented endpoints.
- Drove Agile implementation by leading biweekly standups, sprint planning, and retrospectives across a cross-functional team.
- Tech: Java, TypeScript, Angular, SpringBoot, Maven, Node.js, SwaggerUI

PROJECTS

NoteApp – Full-Stack Note Taking Application Mar 2026 - Apr 2026

- Designing and building a full-stack note-taking app featuring rich text editing, freehand drawing, and PDF annotation layered on top of each other within a single note.
- Implemented JWT-based authentication with BCrypt password hashing, role-based note sharing with READ/EDIT permissions, and a shared notes system with revoke access functionality.
- Architected a clean layered backend with entities, DTOs, repositories, services, and controllers, with a global exception handler for consistent error responses.
- Tech: Angular, TypeScript, Angular Material, Spring Boot (Java), H2/PostgreSQL, Maven, JWT

QR Code Class Attendance – Serverless AWS Attendance System 2025

- Built a serverless QR-based attendance system where teachers generate session-specific QR codes and students scan them to mark attendance, with real-time email notifications via SNS.
- Architected the full AWS infrastructure using Lambda, API Gateway, DynamoDB, S3, CloudFront, and SNS — with a GSI on DynamoDB for efficient session-based attendance queries.
- Hosted the static frontend on S3 with CloudFront for HTTPS support, global caching, and a private bucket configuration.
- Tech: AWS Lambda, API Gateway, DynamoDB, S3, CloudFront, SNS, CloudWatch, JavaScript, HTML/CSS

Cyberspace – End-to-End Encrypted Discussion Platform Aug 2024 - Dec 2024

- Designed and built a fully encrypted chat application from scratch to demonstrate end-to-end security without relying on HTTPS.
- Implemented user authentication with BCrypt password hashing, message verification, and role-based access control across multiple chat boards.
- Architected a three-server system (resource, authentication, and bridge) enabling secure client communication with ECDH key exchange, 2048-bit RSA, and AES-GCM encryption.
- Persisted data using SQLite on both resource and authentication servers for durable session and credential management.
- Tech: React, Node.js, SpringBoot (Java), SQLite, JavaScript, HTML/CSS

TECHNICAL SKILLS

Languages: Java, JavaScript, TypeScript, Python, C, Rust, SQL, R, HTML/CSS, MIPS Assembly

Frameworks & Tools: React, Angular, SpringBoot, Flask, Node.js, Maven, Jinja

Testing & QA: Selenium, JUnit, Cucumber, SpotBugs, Postman, SwaggerUI, Java Path Finder

Infrastructure: AWS (Lambda, DynamoDB, S3, CloudFront, SNS, API Gateway), Linux, Git